

Automated Depth Estimation and 3D Reconstruction of CFRP Subsurface Defects via Optical Pulsed Thermography

Habeeban Memon, Zain Ul Abidin (Corresponding Author), Junaid Ahmed

Abstract—Accurate through-thickness depth prediction of CFRP subsurface delaminations is key for structural assessment, as defects at various ply interfaces have different load-bearing importance. However, while optical pulsed thermography encodes depth in the peak time of the surface thermal response above a defect, conventional regression models often overfit to trained defect shapes and fail on unseen ones. In this paper, we present a two-stage modular pipeline for depth estimation and 3D reconstruction of CFRP subsurface delaminations from raw thermographic data. U-Net, FPN, and DeepLabV3+ segmentation networks are compared based on IoU and Dice scores, and the best-performing architecture is selected for stage 2. Stage 2 utilizes the spatially averaged thermal decay curve and extracts 16 physics-informed features based on the depth and peak time relationship. Regression models are evaluated using specimen-level cross-validation on 6 CFRP specimens, including square defects at 0.2 to 1.0 mm and circular defects at 2.0 to 2.2 mm depth. The best model, utilizing L1 and L2 regularization with column sampling, predicts depth with an MAE of 0.056 mm and an RMSE of 0.085 mm, yielding approximately half the error of the second-best model. These predicted depths and stage 1 masks are used to construct a 3D mesh through Delaunay triangulation within 3 to 5 seconds.

Keywords: CFRP, delamination depth, pulsed thermography, XGBoost, 3D reconstruction, segmentation, spatial overfitting.

I. INTRODUCTION

In safety critical aerospace and renewable energy systems, carbon fiber reinforced polymer (CFRP) composites are the primary structural material. CFRP composites make up roughly half the structural weight of modern aircraft [1], [2]. Wind turbine rotor blades also endure heavy loads for decades [3]. However, the most critical failure mode in these applications is interlaminar delamination. Low energy impacts, like tool drops or bird strikes, cause internal damage without leaving visible marks on the surface [4]. The depth of these hidden defects determines the remaining strength of the structure. Damage at a shallow 0.2 mm depth breaks load transfer paths in the surface layers. Damage at 2.0 mm impacts an entirely different set of layers across the thickness [5]. Therefore, finding only the two dimensional location of a defect is not enough. Without knowing the exact depth, engineers cannot use the inspection data for damage tolerant design or structural health monitoring.

Optical pulsed thermography (OPT) is a non contact inspection method. It uses a high energy flash to heat the specimen surface while an infrared camera records the cooling process [6], [7]. As heat travels into the material, internal defects block the energy because they have lower thermal conductivity. Because of this, the surface directly above a defect cools slower and stays warmer than healthy areas. Using a simple one dimensional heat conduction model, the defect depth d and the peak contrast time t_{peak} are related by [8]:

$$d = \beta \sqrt{\alpha t_{\text{peak}}} \quad (1)$$

where α is the thermal diffusivity and β is a geometric factor. Maldague defined the theoretical principles of this diffusion process [6], building on earlier analytical solutions [8].

Early methods to find depth relied on signal processing of raw thermal data. Shepard et al. showed that thermographic signal reconstruction (TSR) can estimate t_{peak} with less noise by fitting a polynomial to the cooling curve [9]. Schager et al. later expanded this to automate segmentation [10]. Ibarra-Castaneda studied pulsed phase thermography (PPT) to reduce the effects of uneven heating [11]. Rajic introduced principal component thermography (PCT) to improve defect boundary detection [12]. These signal processing techniques align with recent findings in active microwave and optical thermography. These findings strongly suggest using explicit contrast subtraction formulas to measure signal to noise ratio, rather than relying on absolute mean signal measurements [13]. However, the main problem with these physical methods is their need for exact physical calibration [14]. In real composite parts, thermal diffusivity and geometric constants change based on fiber direction, moisture, and curing conditions. Therefore, using equation (1) directly in the field is prone to error without a perfect reference sample.

To avoid strict physical calibration, researchers moved to supervised machine learning. These models learn the relationship between the thermal signal and depth. Peng and Li used a random forest model with extracted temporal and geometric features to achieve a mean absolute error (MAE) of 0.21 mm [15]. Dong et al. proposed 3D residual convolutional neural networks (CNNs) to process the entire thermal cube, achieving an MAE of 0.18 mm [16]. Wei et al. used an instance segmentation network and random forests to predict depths on curved objects [17]. Recent research highlights the success of deep learning for automated composite inspection. These studies emphasize extracting physical features rather than processing raw images [18]–[21]. Dhar et al. showed that 1D CNNs are highly useful for analyzing thermal decay patterns on simulated flat bottom holes [22]. Zheng et al. also validated sequenced thermal signals for direct depth regression [23]. Other groups proposed joint architectures to perform segmentation and depth estimation at the same time. These include networks with residual connections [24], attention U-

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Net models [25], and self training TSR fusion [26], [27]. However, joint networks require paired ground truth images for every sample and are difficult to adjust independently.

Despite these improvements, a major problem remains in current regression models. This problem is spatial overfitting. Experimental datasets usually rely on discrete flat bottom holes or Teflon inserts arranged in rigid grids. Because of this layout, the spatial position of the defect acts as a hidden variable for depth. Instead of learning the heat diffusion physics, unregularized tree models simply memorize the geographic coordinates of the defects. If tested on a specimen with a different geometry, these models fail because they predict depth based on location rather than the thermal curve. Also, trying to fix this by processing the entire thermal volume requires huge GPU memory. Processing scales as $O(HWT)$, which is not practical for high resolution industrial data with thousands of frames [28], [29].

In this paper, we propose a modular, two stage pipeline designed to avoid spatial overfitting and reduce computational loads. First, we segment the defect regions. Then, we extract 16 physics based features from the spatially averaged thermal decay curves of those specific regions. Finally, we regress the depth and triangulate the predictions into an accurate 3D surface mesh. Our novel contributions include:

- 1) An end to end pipeline that localizes defects before regression. This allows the system to process specific physical features instead of massive full volume tensors.
- 2) A robust feature extraction method using 16 specific temporal, statistical, and geometric features derived from the heat conduction equation.
- 3) A comparison of 5 machine learning regression models under strict specimen level cross validation. The best performing model yields a cross validated MAE of 0.056 mm.
- 4) A clear demonstration that standard tree models lose precision due to geometric overfitting, and a regularization strategy to fix it.
- 5) Automated generation of interactive 3D meshed defects through Delaunay triangulation in 3 to 5 seconds per specimen. Defects are modeled accurately as thin, plate like layer delaminations that match the predicted segmentation masks.

A direct numerical comparison with existing deep learning depth estimation models is not possible. There are major differences in experimental hardware setups, flash energy levels, and sampling frequencies across the literature. Also, using classical, unsupervised thermographic methods as a baseline is not appropriate for evaluating a supervised machine learning pipeline. Therefore, this study sets up a strict internal benchmark. We evaluate five different classes of learning algorithms under identical cross validation to isolate algorithmic performance and directly address spatial overfitting.

The remainder of this paper is organized as follows. Section II details proposed methodology for both stage 1 and stage 2. Section III presents the experimental setup. Section IV reports the results and discussion, provides a detailed spatial overfitting

analysis, and demonstrates the 3D reconstruction. Finally, Section V concludes the paper and discusses future work.

II. PROPOSED METHODOLOGY

The proposed pipeline consists of two distinct stages. In the first stage, we perform defect localization using segmentation model inference on the thermographic data. This stage outputs predicted binary defect masks. The second stage then takes these masks as input, calculates physics-based features based on the spatially averaged thermal response of each detected region, and applies depth regression to output a 3D mesh of delaminations reconstructed using Delaunay triangulation.

The overall pipeline execution follows the sequence: data acquisition, segmentation, thermal feature extraction per defect region, depth label prediction, data splitting, regression training and evaluation, 3D reconstruction, and automated spatial localization. Fig. 1 shows the proposed two-stage pipeline for defect depth estimation and 3D reconstruction from raw thermal tensor and Stage 1 segmentation masks.

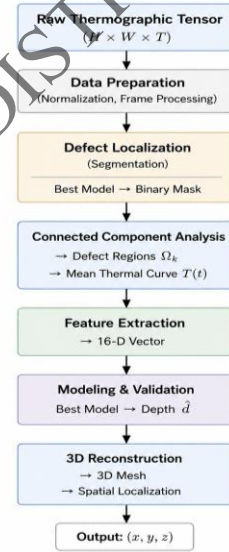


Fig. 1. Proposed two-stage pipeline for defect depth estimation and 3D reconstruction

A. Stage 1: Defect Localization via Segmentation

The primary goal of stage 1 is to determine a predicted binary defect mask $\hat{M} \in \{0,1\}^{H \times W}$ that specifies the region of every subsurface defect. This stage of our pipeline is modular. The masks used for regression in Stage 2 may be generated from raw thermographic image data, derived processed image representations such as PCA or TSR images, or from any multi-channel image from a segmentation pipeline.

Three segmentation model architectures are considered. The first two utilize a ResNet-34 encoder and combine its features with a decoder via either skip-connections (U-Net) or an element-wise top-down addition across feature pyramid levels (FPN) to capture defects at varying scales [30]–[32]. DeepLabV3+ has a ResNet-50 encoder and uses atrous spatial pyramid pooling with various dilation rates to capture multi-scale context at the native image resolution without downsampling [33].

Each architecture is trained using identical parameters: a binary cross-entropy and soft Dice loss, label smoothing of $\varepsilon = 0.1$, an Adam optimizer [34] with a learning rate $\eta_0 = 1 \times 10^{-4}$, and an ImageNet-pretrained encoder. Ground truth binary masks were generated by annotating images from PCA component analysis where defects are most visually distinct. These annotations are utilized solely for the training of the segmentation models and are completely isolated from stage 2 regression.

ALGORITHM 1: SEGMENTATION MODEL TRAINING AND BEST-MASK SELECTION.

1. Input: Raw thermal tensor $X \in \mathbb{R}^{H \times W \times T}$; ground-truth binary masks $M^* \in \{0,1\}^{H \times W}$ for training supervision only.
2. For each architecture $\mathcal{A} \in \{\text{U-Net, FPN, DeepLabV3+}\}$:
 - Train $\mathcal{A}$ on all training-fold specimens using M^* as pixel-level supervision.
3. For inference each trained $\mathcal{A}$, pass the thermographic input for each test specimen to obtain the predicted binary mask: $\hat{M}_{\mathcal{A}} = \mathbb{1}[\mathcal{A}(X) \geq 0.5] \in \{0,1\}^{H \times W}$
4. Compute pixel-level Dice coefficient and IoU for each $\hat{M}_{\mathcal{A}}$ against M^* on test specimens.
5. The best performing architecture is selected as $\mathcal{A}^* = \arg \max_{\mathcal{A}} \text{Dice}(\hat{M}_{\mathcal{A}}, M^*)$
6. Output: Predicted binary mask $\hat{M} \equiv \hat{M}_{\mathcal{A}^*} \in \{0,1\}^{H \times W}$.

B. Stage 2: Feature Extraction, Depth Regression, and Reconstruction

1) Connected Component Extraction and Spatially Averaged Thermal Response

We applied connected component labeling to the predicted mask $\hat{M}$ to find the K individual defects Ω_k . The thermal response is spatially averaged over the pixels that make up Ω_k for each time frame t to mitigate pixel-level camera noise:

$$T_k(t) = \frac{1}{|\Omega_k|} \sum_{(i,j) \in \Omega_k} X_{\text{norm}}(i,j,t) \quad (2)$$

where $T_k(t)$ is the spatially averaged normalized temperature for defect region k . The globally normalized thermal intensity is given by $X_{\text{norm}}(i,j,t) = X(i,j,t) / \max_{i,j,t} X(i,j,t)$. Global normalization is strictly used to preserve the relative shape of the temporal decay without altering the inherent physical timing. Averaging over space suppresses pixel-level noise proportionally to $1/\sqrt{|\Omega_k|}$ to recover a smoothed mean decay representing the physical response of the defect's thermal signal. Mean Thermal Curves are shown in Fig. 2.

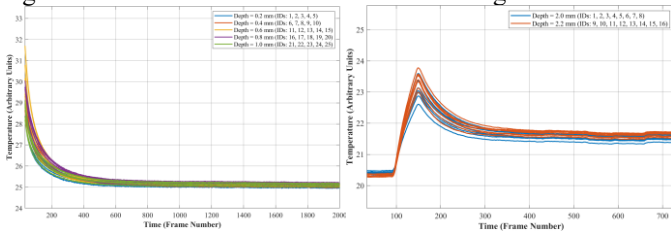


Fig. 2. Mean thermal curves T_t for defects in SQ_01 (left) and CI_01 (right)

2) Feature Engineering

Sixteen features derived from the mean thermal curve $T_k(t)$ and the defect geometry are presented in Tables I, II, III, and IV. We define three intermediate quantities that are shared between features:

$$\Delta T_k(t) = T_k(t+1) - T_k(t) \quad (3)$$

where $\Delta T_k(t)$ represents the inter-frame change in temperature.

$$\mu_T = \frac{1}{T} \sum_{t=1}^T T_k(t) \quad (4)$$

T is the total number of frames in the acquisition.

$$\tilde{T}_k(t) = T_k(t) - \mu_T \quad (5)$$

where $\tilde{T}_k(t)$ represents the zero-mean centered curve with μ_T defined in equation (4).

The sixteen features are categorized into four groups based on the underlying physics they represent.

Temporal features reflect the time-dependent aspects of the decay process. t_{pk} is the most physics-aligned feature, having a direct relationship with depth via equation (1). The linearized depth proxy $d_{\text{rel}} = \sqrt{t_{pk}}$ is derived from rearranging equation (1).

TABLE I. TEMPORAL FEATURES EXTRACTED FROM THE MEAN THERMAL CURVE $T_k(t)$.

Feature	Formula	Physical Interpretation
t_{pk}	$\arg \max_t T_k(t),$ $t \in \{1, \dots, T\}$	Time to peak; primary depth proxy via (1)
τ_r	$t_{pk} - t_{\text{onset}}$	Rise duration; correlated with t_{pk} and defect proximity
T_{max}	$\max_t T_k(t)$	Peak surface temperature; inversely related to depth
$\dot{T}_{\text{max}}$	$\max_t \Delta T_k(t) / \Delta t$	Maximum heating rate; sensitive to defect proximity
d_{rel}	$\sqrt{t_{pk}}$	Linearized depth proxy from rearrangement of (1)

Energy-based features reflect the shape of the curve integral, indicating how much heat the defect retains during the entire acquisition time.

TABLE II. ENERGY-BASED FEATURES EXTRACTED FROM THE MEAN THERMAL CURVE.

Feature	Formula	Physical Interpretation
A_{auc}	$\sum_{t=1}^T T_k(t) \Delta t$	Area under curve; total retained thermal energy
E	$\sum_{t=1}^T T_k(t)^2 \Delta t$	Signal energy; depth-dependent through decay rate
T_{tail}	mean $T_k(t),$ $t \in [0.8T, T]$	Late-time mean temperature; sensitive to deep defects

Statistical features are independent of absolute temperature and provide a description of the decay shape. Skewness and kurtosis indicate how much the decay curve is shifted to the right or how heavy-tailed it is, respectively. The lag-1 autocorrelation indicates the smoothness of the decay.

TABLE III. STATISTICAL FEATURES EXTRACTED FROM THE MEAN THERMAL CURVE.

Feature	Formula	Physical Interpretation
σ_T	$\sqrt{\frac{1}{T} \sum_t \tilde{T}_k(t)^2}$	Standard deviation; spread of thermal response
γ_1	$\frac{1}{T\sigma_T^3} \sum_t \tilde{T}_k(t)^3$	Skewness; asymmetry of the decay profile
γ_2	$\frac{1}{T\sigma_T^4} \sum_t \tilde{T}_k(t)^4 - 3$	Excess kurtosis; tail weight of the decay curve
ρ	$\frac{\sum_t \tilde{T}_k(t) \tilde{T}_k(t+1)}{\sum_t \tilde{T}_k(t)^2}$	Lag-1 autocorrelation; temporal smoothness of decay

Finally, the geometric features take the predicted defect shape into account and serve as an indicator of when the one-dimensional diffusion model may not accurately predict depth due to significant lateral heat diffusion (wide, shallow defects). The equivalent circular diameter and aspect ratio are derived from the bounding box of the defect region. The parameter β in equation (1) should account for the geometric factors which impact the speed and nature of thermal diffusion. The linked feature $d_{rel,g} = \sqrt{t_{pk}}$ passes the peak-time depth estimate into the geometry group.

TABLE IV. GEOMETRIC FEATURES EXTRACTED FROM THE DEFECT REGION Ω_k .

Feature	Formula	Physical Interpretation
A_k	$ \Omega_k $	Area in pixels
D_k	$2\sqrt{A_k/\pi}$	Equivalent circular diameter
AR_k	W_{bb}/H_{bb}	Aspect ratio; proxy for defect shape and β correction
$d_{rel,g}$	$\sqrt{t_{pk}}$	Peak-time depth proxy passed as cross-group linking feature

3) Machine Learning Regressors

Regression models were evaluated on the extracted features using Random Forest (RF), Gradient Boosting (GBM), Support Vector Regression (SVR), XGBoost, and an advanced MLP (Adv-MLP). RF and GBM are tree-based ensemble methods [35], [36], while SVR employs an RBF-kernel formulation for nonlinear regression. XGBoost is a regularized gradient boosting framework designed for efficient and robust prediction [37]. Adv-MLP is a feedforward neural network with three hidden layers (128–64–32), enhanced with batch normalization and dropout, and trained using Huber loss with Adam optimization and early stopping based on validation error.

ALGORITHM 2: PHYSICS-INFORMED FEATURE EXTRACTION AND DEPTH REGRESSION.

1. **Input:** Predicted binary mask $\hat{M} \in \{0,1\}^{H \times W}$ from Stage 1; globally normalized thermal tensor $X_{norm} \in \mathbb{R}^{H \times W \times T}$; ground-truth depth labels $\{d_k^{true}\}$ for training and evaluation only.
2. Apply connected component labeling to $\hat{M}$ to identify K distinct defect regions Ω_k , $k = 1, \dots, K$.
3. For each region Ω_k and each frame t , compute the spatially averaged thermal curve $T_k(t)$.
4. Assign ground-truth labels d_k^{true} using specimen geometry.
5. Standardize all features using training-set statistics only.

6. Train all five regression models on the standardized training feature vectors and depth labels.
7. For each test defect k , predict depth as $\hat{d}_k = f_{best}(\hat{f}_k)$.
8. Compute MAE, RMSE, and R^2 on test defects.
9. **Output:** Predicted depth $\hat{d}_k$ for each detected defect $k = 1, \dots, K$.

C. Three-Dimensional Reconstruction and Automated Spatial Localization

Each defect Ω_k is treated as a thin-layer delamination with depth $\hat{d}_k$ (predicted by the best-performing regression model) and represented by top and bottom surfaces. Pixel coordinates are converted to physical units via a spatial calibration factor:

$$x_{mm} = j \cdot s_{px}, y_{mm} = i \cdot s_{px}, s_{px} = \frac{L_{spec}}{N_{px}} \quad (9)$$

where j and i are pixel coordinates (column and row, respectively), s_{px} is the pixel size in mm/pixel (e.g., 0.586 mm/pixel for Dataset 1's 512-pixels over a 300 mm edge), and L_{spec} and N_{px} are the specimen edge length and number of pixels, respectively. For Dataset 2, $s_{px} = 0.391$ mm/pixel. The delamination surfaces are assigned fixed Z coordinates:

$$Z_{top}(x, y) = \hat{d}_k \forall (x, y) \in \Omega_k \quad (10)$$

$$Z_{bot}(x, y) = \hat{d}_k + \delta_{gap} \forall (x, y) \in \Omega_k \quad (11)$$

where $\hat{d}_k$ is the predicted depth, and $\delta_{gap} = 0.0075$ mm is the Teflon insert thickness. Two point clouds (top and bottom) are generated for each defect:

$$p_{ij}^{top} = (j \cdot s_{px}, i \cdot s_{px}, \hat{d}_k) \quad (12)$$

$$p_{ij}^{bot} = (j \cdot s_{px}, i \cdot s_{px}, \hat{d}_k + \delta_{gap}) \quad (13)$$

The Delaunay triangulation algorithm [38] is applied independently to the xy -coordinates of each point cloud to create triangulated surface meshes. The mean surface distance (MSD) for each defect:

$$MSD_k = |\hat{d}_k - d_k^{true}| \quad (14)$$

where d_k^{true} is the nominal ground-truth depth. The MSD is equal to the regression MAE per defect because each pixel within Ω_k is assigned the same predicted depth. The 3D centroid position (automated spatial localization) of the defect:

$$\mathbf{c}_k^{3D} = (x_k \cdot s_{px}, y_k \cdot s_{px}, \hat{d}_k) \quad (15)$$

where x_k and y_k are the centroid pixel coordinates from Stage 1, s_{px} is the physical pixel size, and $\hat{d}_k$ is the predicted depth. This gives the precise 3D location of the defect for integration into digital twins or design models.

ALGORITHM 3: THREE-DIMENSIONAL RECONSTRUCTION AND AUTOMATED SPATIAL LOCALIZATION.

1. **Input:** Predicted binary mask $\hat{M} \in \{0,1\}^{H \times W}$ from Stage 1; predicted depths $\{\hat{d}_k\}_{k=1}^K$ from Algorithm 2; pixel scale s_{px} ; nominal insert thickness $\delta_{gap} = 0.0075$ mm.
2. For each pixel $(i, j) \in \Omega_k$, compute physical coordinates (x_{mm}, y_{mm}) .

3. For each defect region Ω_k , construct top-surface point cloud P_k^{top} and bottom-surface point cloud P_k^{bot} .
4. Apply Delaunay triangulation to the xy -coordinates of P_k^{top} and P_k^{bot} independently.
5. Compute the three-dimensional centroid position c_k^{3D} for each defect.
6. For each defect k , compute MSD_k .
7. **Output:** Per-defect triangulated mesh $\mathcal{M}_k$, 3D centroid c_k^{3D} , equivalent diameter D_k^{phys} , volume V_k , and MSD_k .

III. EXPERIMENTAL SETUP

A. Experimental Thermography Datasets

To evaluate the proposed framework, thermographic image sequences from six CFRP specimens obtained from two experimental datasets are utilized. The datasets were acquired using optical pulsed thermography (OPT), a non-contact inspection technique widely used for the detection of subsurface defects in composite materials. In OPT, thermal excitation and infrared data acquisition are performed on the same side of the specimen in a reflection-mode configuration, enabling single-sided inspection of composite structures [6], [7]. Dataset 1 is from Erazo-Aux et al. [39] comprises three CFRP specimens, labelled SQ_01 (flat), SQ_02 (curved), and SQ_03 (trapezoidal) which correspond to specimens CFRP006, CFRP007 and CFRP008 in the original publication. Dataset 2, compiled by Ahmed et al. [40], involves three CFRP specimens, labelled CI_01, CI_02, and CI_03 which correspond to specimens 3, 1 and 2 in the original publication.

B. Assignment of Depth Labels

For training and evaluation only, depth labels are derived from the physical sample geometry. In square-insert specimens, defects are arranged column-wise, and depth is assigned according to its column index from a predefined depth map $\{1.0, 0.6, 0.2, 0.4, 0.8\}$ mm [39]. In circular-insert specimens, defects are arranged row-wise, and depth is assigned based on the row index corresponding to $\{2.0, 2.2\}$ mm [40].

C. Cross-Validation Partitioning

All predictions are made at the defect level (regression). The specimen-level three-fold cross-validation is maintained for both pipeline steps to avoid data leakage between training, validation, and testing. One square-insert and one circular-insert specimen are assigned to the training, validation, and test sets for each fold. The full partitioning is detailed in Table V.

TABLE V. THREE-FOLD SPECIMEN-LEVEL CROSS-VALIDATION PARTITIONING.

Fold	Training Specimens	Validation Specimens	Test Specimens
1	SQ_01, CI_01 (41: 25 SQ + 16 CI)	SQ_02, CI_02 (35: 25 SQ + 10 CI)	SQ_03, CI_03 (31: 25 SQ + 6 CI)
2	SQ_03, CI_03 (31: 25 SQ + 6 CI)	SQ_01, CI_01 (41: 25 SQ + 16 CI)	SQ_02, CI_02 (35: 25 SQ + 10 CI)
3	SQ_02, CI_02 (35: 25 SQ + 10 CI)	SQ_03, CI_03 (31: 25 SQ + 6 CI)	SQ_01, CI_01 (41: 25 SQ + 16 CI)

D. Evaluation Metrics

Three evaluation metrics were calculated. Mean absolute error:

$$\text{MAE} = \frac{1}{N} \sum_{n=1}^N |d_n - \hat{d}_n| \quad (16)$$

where d_n is the true defect depth, $\hat{d}_n$ is the predicted defect depth, and N is the number of test defects. MAE is the primary metric because it corresponds to an interpretable physical unit (millimeters). Root mean square error:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{n=1}^N (d_n - \hat{d}_n)^2} \quad (17)$$

RMSE is sensitive to individual large errors and helps diagnose potential model collapse on certain samples. Coefficient of determination:

$$R^2 = 1 - \frac{\sum_{n=1}^N (d_n - \hat{d}_n)^2}{\sum_{n=1}^N (d_n - \bar{d})^2} \quad (18)$$

where $\bar{d} = \frac{1}{N} \sum_{n=1}^N d_n$ is the mean true depth. R^2 is reported for completeness but is less meaningful for circular specimens where the label distribution has very low variance. The denominator can be much smaller than the numerator, resulting in a negative R^2 for models that do not perform substantially better than simply predicting the mean.

IV. RESULTS AND DISCUSSION

A. Segmentation Performance and Mask Quality

Before evaluating depth regression, we must assess the quality of the Stage 1 predicted masks $\hat{M}$. Any insufficient detection or boundary error in $\hat{M}$ transfers directly into the Stage 2 feature extraction. All three network architectures were tested on the held-out test specimens for each fold. The results are summarized in Table VI.

TABLE VI. SEGMENTATION RESULTS FOR FOLD 1, FOLD 2 AND FOLD 3

Specimens	U-Net		FPN		DeepLabV3+	
	Dice	IoU	Dice	IoU	Dice	IoU
SQ_03	0.84	0.72	0.85	0.73	0.83	0.72
SQ_02	0.84	0.72	0.85	0.75	0.86	0.76
SQ_01	0.86	0.75	0.85	0.74	0.84	0.73
CI_03	0.34	0.20	0.39	0.24	0.27	0.16
CI_02	0.52	0.35	0.31	0.18	0.44	0.28
CI_01	0.26	0.15	0.19	0.10	0.20	0.11

Across all three folds, the Dice scores for square specimens ranged between 0.83 and 0.87, showing that the pipeline can reliably detect shallow square inserts and map their boundaries. Dice scores for circular specimens varied between 0.19 and 0.52 depending on the fold and architecture. Based on a consistent mean IoU across all datasets, DeepLabV3+ was selected as the final mask-generating model. Its atrous spatial pyramid pooling module aggregates context over multiple scales, which successfully suppresses background noise for both square and circular defects.

The models successfully detected 100% of the actual defects in all square specimens as shown in Fig. 3. For circular

specimens, detection varied by size: CI_03 (Fold 1) had all six defects detected, CI_02 (Fold 2) had eight out of ten detected, and CI_01 (Fold 3) had nine out of sixteen detected. The undetected circular defects were primarily the 2 mm diameter inserts, which sit closely together and fall near or below the uncooled camera's spatial resolution limit of 640×480 pixels.

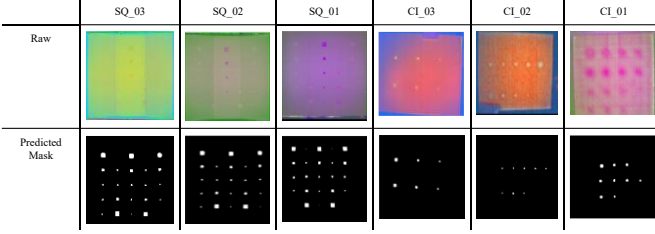


Fig. 3. Predicted binary masks from the best-performing segmentation architecture for each of the six specimens.

B. Depth Regression Results

1) Fold 1 Performance

In Fold 1, the training set contained defects from SQ_01 and CI_01, the validation set contained SQ_02 and CI_02, and the test set consisted of SQ_03 (25 square defects) and CI_03 (6 circular defects). The overall depth regression results are provided in Table VII.

TABLE VII. DEPTH REGRESSION RESULTS FOR FOLD 1 ON TEST SPECIMENS SQ_03 AND CI_03.

Model	Specimen	MAE (mm)	RMSE (mm)	R^2
Random Forest	SQ_03	0.0102	0.0171	0.9963
	CI_03	0.0647	0.1021	-0.042
	Overall	0.0207	0.0475	0.9946
Gradient Boosting	SQ_03	0.0125	0.0184	0.9958
	CI_03	0.0400	0.0861	0.259
	Overall	0.0178	0.0413	0.9959
SVR	SQ_03	0.1331	0.1692	0.642
	CI_03	0.1321	0.1712	-1.930
	Overall	0.1329	0.1696	0.931
XGBoost	SQ_03	0.0099	0.0169	0.9964
	CI_03	0.1154	0.1309	-0.714
	Overall	0.0303	0.0596	0.9915
Adv-MLP	SQ_03	0.1458	0.1725	0.628
	CI_03	0.3385	0.3665	-12.43
	Overall	0.1831	0.2236	0.880

Gradient Boosting yielded the best overall results for Fold 1 with an MAE of 0.018 mm, while XGBoost performed second with an MAE of 0.030 mm. Negative R^2 values for some models on CI_03 do not represent poor performance, but rather occur because the variance in actual depth for this specific circular specimen is extremely small (only 2.0 or 2.2 mm). The Adv-MLP network produced an overall MAE of 0.183 mm. Because the training set contains only 41 discrete defects, the large parameter network simply memorized the training set instead of learning the physical relationship between thermal decay and depth.

2) Fold 2 Performance (Spatial Overfitting Analysis)

Fold 2 provides the most critical diagnostic insight. The training set (SQ_03) contains 25 defects arranged in a rigid 5-column array, meaning depth directly correlates with spatial position. We tested this model on SQ_02 (curved plate) and

CI_02 (circular defects with no column structure). The regression results for Fold 2 are detailed in Table VIII.

TABLE VIII. DEPTH REGRESSION RESULTS FOR FOLD 2 ON TEST SPECIMENS SQ_02 AND CI_02.

Model	Specimen	MAE (mm)	RMSE (mm)	R^2
Random Forest	SQ_02	0.0120	0.0158	0.9969
	CI_02	0.5209	0.5326	-27.37
	Overall	0.1574	0.2850	0.844
Gradient Boosting	SQ_02	0.0162	0.0199	0.9951
	CI_02	0.6834	0.7232	-51.29
	Overall	0.2068	0.3869	0.712
SVR	SQ_02	0.3476	0.3844	-0.847
	CI_02	0.1254	0.1363	-0.857
	Overall	0.2841	0.3330	0.786
XGBoost	SQ_02	0.0359	0.0461	0.9734
	CI_02	0.0756	0.0967	0.066
	Overall	0.0473	0.0647	0.992
Adv-MLP	SQ_02	0.5221	0.5730	-3.103
	CI_02	0.1370	0.1622	-1.632
	Overall	0.4121	0.4919	0.534

Random Forest and Gradient Boosting failed dramatically on CI_02, returning MAE values of 0.521 mm and 0.683 mm, respectively. This error is larger than the total depth range of the specimen, indicating a total loss of physical calibration. The tree models used the 5-column structure of the training data as a spatial proxy for depth. When tested on the circular specimen where spatial position means nothing, they failed. This clearly demonstrates spatial overfitting as described by Breiman [35].

XGBoost successfully avoided this collapse. Its L1 and L2 regularization terms combined with column subsampling prevented the model from memorizing geometric structures. XGBoost generalized perfectly to CI_02, returning an overall MAE of 0.047 mm for Fold 2 by relying on the physically meaningful temporal feature $d_{\text{rel}} = \sqrt{t_{pk}}$ rather than spatial location.

3) Fold 3 Performance

The training set consists of defects from SQ_02 and CI_02 and the validation set contains defects from SQ_03 and CI_03. The test set consists of SQ_01 (25 square defects, 0.2–1.0 mm depth) and CI_01 (16 circular defects, 2–8 mm diameter, 2.0–2.2 mm depth). The regression results are detailed in Table IX.

TABLE IX. DEPTH REGRESSION RESULTS FOR FOLD 3 ON TEST SPECIMENS SQ_01 AND CI_01.

Model	Specimen	MAE (mm)	RMSE (mm)	R^2
Random Forest	SQ_01	0.1751	0.2437	0.257
	CI_01	0.0362	0.0414	0.829
	Overall	0.1209	0.1921	0.937
Gradient Boosting	SQ_01	0.2215	0.2812	0.011
	CI_01	0.0260	0.0292	0.915
	Overall	0.1452	0.2204	0.917
SVR	SQ_01	0.2288	0.2692	0.094
	CI_01	0.3870	0.3987	-14.90
	Overall	0.2905	0.3259	0.819
XGBoost	SQ_01	0.1192	0.1608	0.677
	CI_01	0.0414	0.0527	0.722
	Overall	0.0888	0.1298	0.971
Adv-MLP	SQ_01	0.3899	0.4293	-1.303
	CI_01	0.8308	0.8356	-68.82
	Overall	0.5620	0.6204	0.346

GBoost again recorded the lowest overall MAE at 0.089 mm. All models struggled slightly on the SQ_01 square specimen compared to previous folds. This is due to surface geometry mismatch: the models trained on a curved plate (SQ_02) and tested on a flat plate (SQ_01). Altering the surface curvature shifts the spatial profile of the heat distribution, changing the t_{peak} at every depth column. Despite this, the regularized models still outperformed the unregularized methods significantly.

C. Three-Dimensional Reconstruction Results

In the full 3D reconstruction of SQ_03 (Fig. 4), the surfaces of the predicted and ground truth delamination layers align almost perfectly at shallow depths (0.2 and 0.4 mm) and show very minimal separation at 1.0 mm. The reconstruction of CI_03 (Fig. 5) clearly resolves the 0.2 mm spacing between the two levels of circular defects.

Because we process the defects based on segmented regions and assign a constant predicted depth to all points within a specific defect Ω_k the Delaunay triangulation introduces zero geometric error. The final mean surface distance (MSD) exactly equals the regression MAE for each defect. By processing highly localized physics features rather than massive full-volume tensors, the full 3D reconstruction completed in just 3 to 5 seconds per specimen on standard workstation hardware.

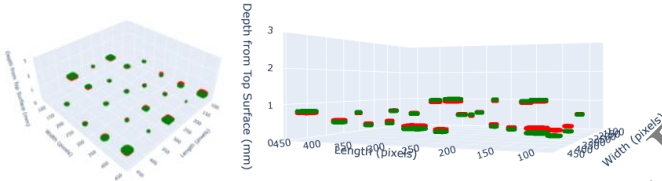


Fig. 4. 3D reconstruction of SQ_03 (Fold 1 test specimen),

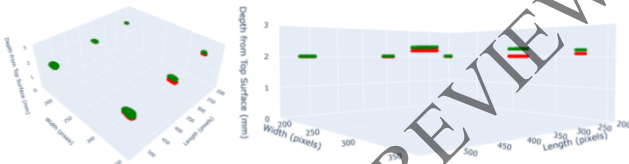


Fig. 5. 3D reconstruction of CI_03 (Fold 1 test specimen)

V. CONCLUSION

This paper demonstrates a modular two-stage pipeline for physics-informed depth estimation and 3D reconstruction of subsurface delaminations in CFRP composites. In Stage 1, we evaluated multiple segmentation architectures and selected DeepLabV3+ based on robust IoU and Dice scores. In Stage 2, we extracted physically meaningful features from spatially averaged thermal curves to train regression models under strict cross-validation. Our analysis proved that unregularized tree-based models suffer heavily from spatial overfitting when trained on geometric defect grids. By employing XGBoost with L1 and L2 regularization, we successfully forced the model to learn the underlying heat diffusion physics, yielding an overall MAE of 0.056 mm and an RMSE of 0.085 mm. The pipeline outputs an accurate 3D mesh via Delaunay triangulation in just 3 to 5 seconds per specimen.

Several limitations remain. The current framework assumes delamination depth is spatially invariant across a single defect and cannot yet map sloped or impact-induced multi-level delaminations. Furthermore, very small defects near the spatial resolution limit of the camera remain difficult to detect, and any segmentation boundary errors naturally propagate into the depth estimation features. Future work will focus on expanding this methodology to perform per-pixel depth estimation for arbitrarily shaped defects without succumbing to spatial overfitting, alongside testing the framework on real-world impact damage.

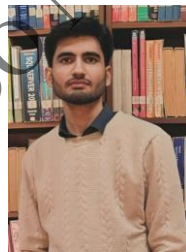
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